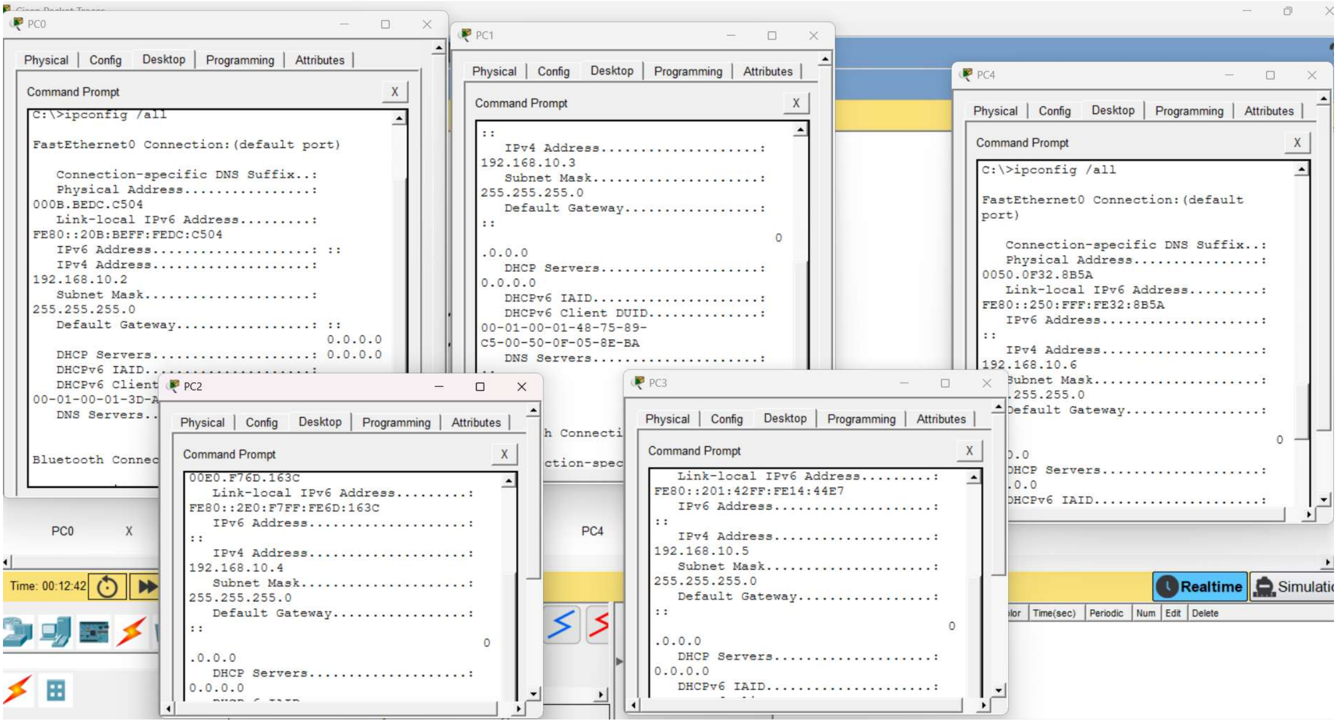
Day 1-5 Build and configure a small office network

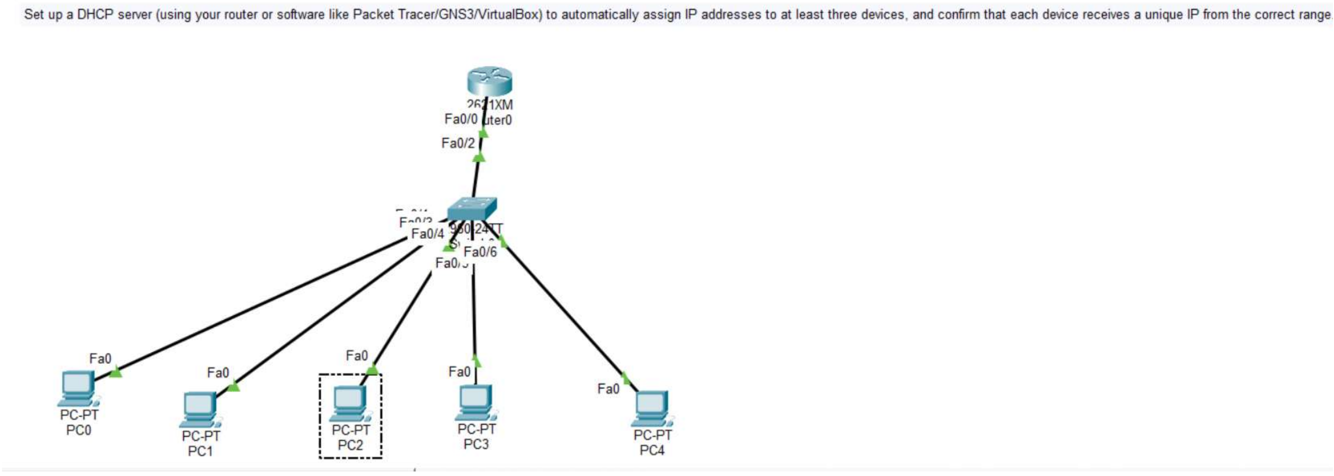
Task 1: Assign static IP addresses to five devices (laptops or desktops) on your home or lab network, ensuring each is in the same subnet, and verify connectivity between them using the ping command.




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time=2ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
Reply from 192.168.10.3: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

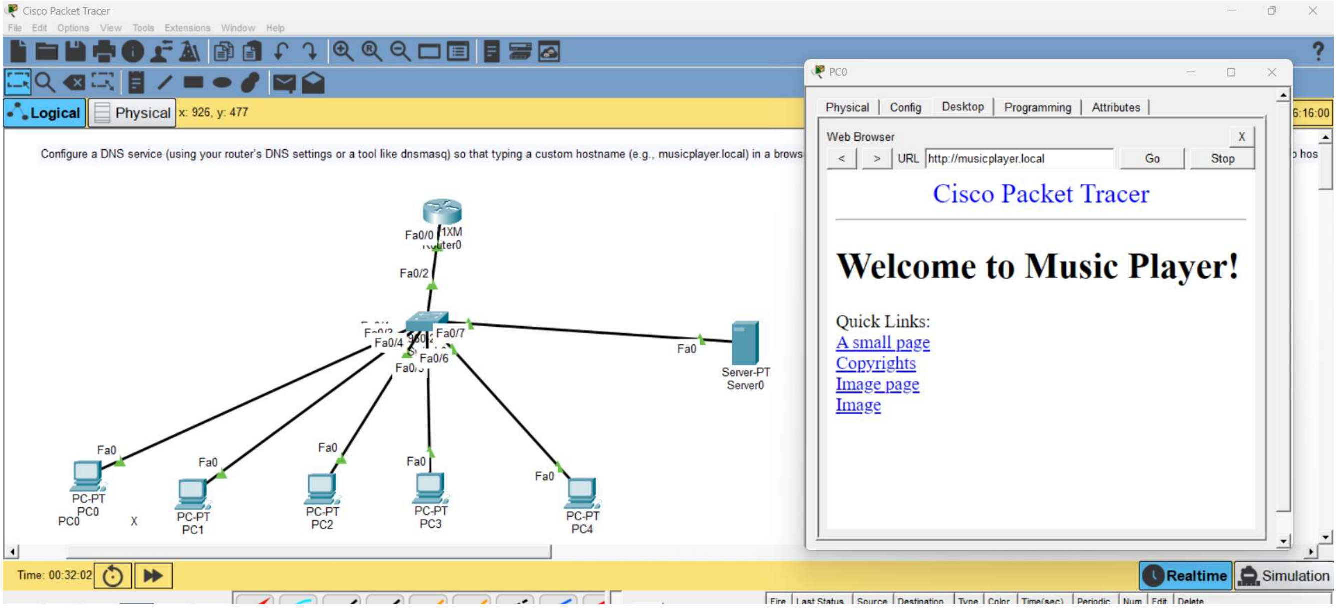
```
C:\>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:

Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128
Reply from 192.168.10.4: bytes=32 time<1ms TTL=128
Reply from 192.168.10.4: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.10.5

Pinging 192.168.10.5 with 32 bytes of data:

Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128
Reply from 192.168.10.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Task 2: Set up a DHCP server (using your router or software like Packet Tracer/GNS3/VirtualBox) to automatically assign IP addresses to at least three devices, and confirm that each device receives a unique IP from the correct range.

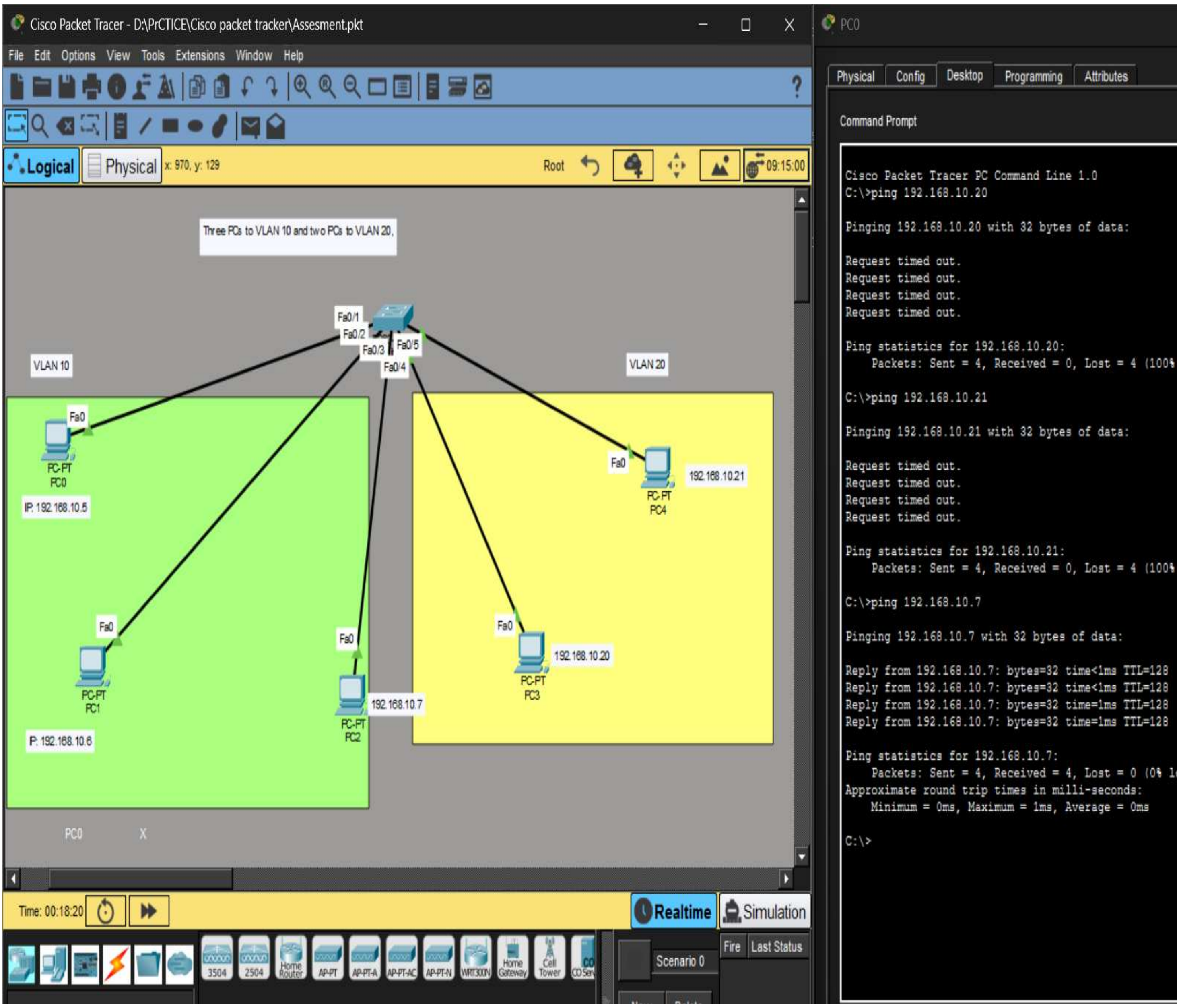



```
  <cr>
Router#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.10.2     000B.BEDC.C504           --                     Automatic
192.168.10.3     0050.0F05.8EBA           --                     Automatic
192.168.10.4     00E0.F76D.163C           --                     Automatic
192.168.10.5     0001.4214.44E7           --                     Automatic
192.168.10.6     0050.0F32.8B5A           --                     Automatic
Router#
```

Task 3: Configure a DNS service (using your router's DNS settings or a tool like dnsmasq) so that typing a custom hostname (e.g., musicplayer.local) in a browser on any connected device opens a specific local web page or device interface. Hint: You can map hostnames to local IPs using the DNS service or by editing the hosts file for a quick test.



Task 4: Create two VLANs (e.g., VLAN 10: 'Workstations', VLAN 20: 'Smart Devices') on your switch or simulation tool, assign at least two devices to each VLAN, and demonstrate that devices in the same VLAN can communicate, but not with devices in the other VLAN.Hint: Use ping or file sharing to test connectivity between devices in different VLANs.


Cisco Packet Tracer - D:\PrCTICE\Cisco packet tracker\Assesment.pkt
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 970, y: 129
Root
09:15:00
Three PCs to VLAN 10 and two PCs to VLAN 20.
Fa0/1
Fa0/2
Fa0/3
Fa0/5
Fa0/4
VLAN 10
VLAN 20
Fa0
PC-PT
PC0
IP: 192.168.10.5
PC-PT
PC1
IP: 192.168.10.6
PC-PT
PC2
192.168.10.7
PC-PT
PC3
192.168.10.20
PC-PT
PC4
192.168.10.21
PC0
X
Time: 00:18:20
Realtime
Simulation
Scenario 0
Fire
Last Status
PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.20
Pinging 192.168.10.20 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.10.20:
Packets: Sent = 4, Received = 0, Lost = 4 (100%
C:\>ping 192.168.10.21
Pinging 192.168.10.21 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.10.21:
Packets: Sent = 4, Received = 0, Lost = 4 (100%
C:\>ping 192.168.10.7
Pinging 192.168.10.7 with 32 bytes of data:
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time<1ms TTL=128
Reply from 192.168.10.7: bytes=32 time=1ms TTL=128
Reply from 192.168.10.7: bytes=32 time=1ms TTL=128
Ping statistics for 192.168.10.7:
Packets: Sent = 4, Received = 4, Lost = 0 (0% l
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

Task 5: Simulate internet access for your LAN by configuring a router with NAT (Network Address Translation), then test that devices on your LAN can access an external site (e.g., www.wikipedia.org) or a simulated external server.

```
% Unrecognized command
Router#show ip nat ?
  statistics    Translation statistics
  translations  Translation entries
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 20.0.0.100:1      10.0.0.2:1         30.0.0.3:1         30.0.0.3:1
icmp 20.0.0.100:2      10.0.0.2:2         30.0.0.3:2         30.0.0.3:2
icmp 20.0.0.100:3      10.0.0.2:3         30.0.0.3:3         30.0.0.3:3
icmp 20.0.0.100:4      10.0.0.2:4         30.0.0.3:4         30.0.0.3:4
icmp 20.0.0.150:1      10.0.0.4:1         30.0.0.2:1         30.0.0.2:1
icmp 20.0.0.150:2      10.0.0.4:2         30.0.0.2:2         30.0.0.2:2
icmp 20.0.0.150:3      10.0.0.4:3         30.0.0.2:3         30.0.0.2:3
icmp 20.0.0.150:4      10.0.0.4:4         30.0.0.2:4         30.0.0.2:4
---  20.0.0.100        10.0.0.2           ---                ---
tcp 172.1.1.1:8443     192.168.3.2:8443   ---                ---

Router#
```


```
Router(config)#do show ip nat translations
Pro  Inside global       Inside local       Outside local      Outside global
udp 20.0.0.150:1025     10.0.0.4:1025      30.0.0.3:53        30.0.0.3:53
udp 20.0.0.150:1026     10.0.0.4:1026      30.0.0.3:53        30.0.0.3:53
udp 20.0.0.150:1027     10.0.0.4:1027      30.0.0.3:53        30.0.0.3:53
udp 20.0.0.150:1028     10.0.0.4:1028      30.0.0.3:53        30.0.0.3:53
udp 20.0.0.150:1029     10.0.0.4:1029      30.0.0.3:53        30.0.0.3:53
udp 20.0.0.150:1030     10.0.0.4:1030      30.0.0.3:53        30.0.0.3:53
udp 20.0.0.150:1031     10.0.0.4:1031      30.0.0.3:53        30.0.0.3:53
---  20.0.0.100         10.0.0.2           ---                ---
tcp 172.1.1.1:8443      192.168.3.2:8443   ---                ---
tcp 20.0.0.150:1025     10.0.0.4:1025      30.0.0.3:80        30.0.0.3:80
tcp 20.0.0.150:1026     10.0.0.4:1026      30.0.0.3:80        30.0.0.3:80
tcp 20.0.0.150:1027     10.0.0.4:1027      30.0.0.3:80        30.0.0.3:80
tcp 20.0.0.150:1028     10.0.0.4:1028      30.0.0.3:80        30.0.0.3:80

Router(config)#
```

DNS Resolution Success:

- **`udp ... 30.0.0.3:53` entries:** Your client (`10.0.0.4`) sent DNS lookup requests to Server0 (`30.0.0.3`) on port 53 to resolve `www.wikipedia.org`.
- **NAT Translation Active:**
- The private local address (`10.0.0.4`) was dynamically translated to the public address (`20.0.0.150`) on Router0 before traveling across the WAN link.
- **End-to-End Simulation:**
- The web page "`Welcome to Wikipedia! Simulated external website via NAT and DNS.`" loaded seamlessly in PC0's web browser using the domain name.